

Ripper User Manual

Landon Viator, Viator DSP

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1 Ripper Overview



Figure 1: Ripper

Ripper is a versatile distortion plugin designed as a free, sonically-rich alternative to popular analog-style distortion tools. It combines multiple distortion models with flexible tone shaping and dynamic control, allowing anything from subtle harmonic enhancement to aggressive saturation and sound design.

At its core, Ripper provides a Drive control to push the signal into distortion and a Mix control for parallel processing, making it easy to blend distortion with the original signal to achieve subtle effects. A Tone knob controlling a tilt filter allows quick adjustment of overall brightness or weight, while dedicated high-pass and low-pass filters enable precise control over the affected frequency range.

An Analog mode switches the filters into Moog-style ladder filters, which introduces additional analog character for a more classic feel. The Rip button applies a fixed +15 dB input boost, instantly driving the signal harder into the distortion stages for more extreme effects.

A built-in meter displays the amount of distortion in decibels, which provides clear visual feedback on how much the signal is being distorted. This makes it easy to dial in consistent results and compare settings on different sources.

Ripper is suitable for a wide range of applications, including drums, guitars, bass, synths, and full mixes, and is designed to be both straightforward for quick results and flexible enough for more experimental or

fine-tuned use.

2 Controls

2.1 Drive Knob

Controls the amount of signal fed into the distortion model. Higher values increase harmonic content, while lower values provide subtle saturation.

2.2 Mix Knob

Controls the balance between the processed signal and the original dry signal. Lower values blend in subtle saturation, while higher values apply the distortion more fully.

2.3 Tone Knob

Adjusts the overall tonal balance using a tilt filter, simultaneously boosting one end of the spectrum while attenuating the other. Each distortion model uses a different pivot frequency, which has been fine-tuned so that the tonal effect interacts naturally with the character of that model.

2.4 High-Pass Knob

Removes low-frequency content from the signal before it reaches the distortion stage, with a cutoff range from 20 Hz to 1000 Hz. Raising the cutoff tightens the low end and prevents excessive low-frequency distortion.

2.5 Low-Pass

Removes high-frequency content from the signal after the distortion stage, with a cutoff range from 1 kHz to 20 kHz. Lowering the cutoff

smooths harsh harmonics and tames high-frequency artifacts introduced by distortion, making the result cleaner and more controlled.

2.6 Rip Button

Applies a fixed +15 dB input boost before distortion, allowing the signal to be driven aggressively into high-gain distortion.

2.7 Analog Button

Switches the high-pass and low-pass filters from standard linear filters to Moog-style ladder filters.

2.8 Distortion Type Knob

Ripper contains six distortion models to choose from: Class B Amplifier, Tube, Circle Map, Tape, Overdrive, and Bit Reducer. Use the Distortion Type knob to switch the active model.



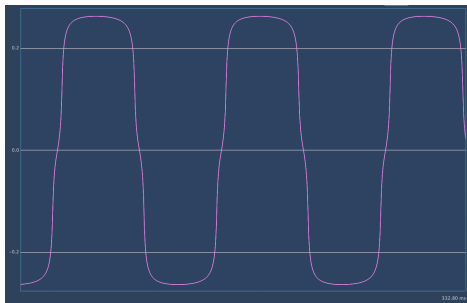
Figure 2: Distortion Type Knob

The following subsections describe the inner workings of each distortion model and the sonic characteristics.

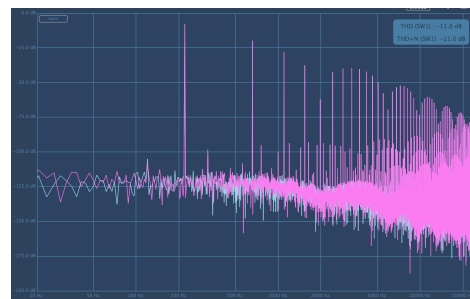
2.9 Class B Amplifier

The Class-B amplifier model is based on research by Will Pirkle, derived from his analysis and interpretation of various vacuum-tube and amplifier patents [1].

The Class-B amplifier is composed of three key elements: a softly squared-off waveform, natural tube-style compression, and crossover distortion produced by separate asymmetric and symmetric wave-shapers applied to the positive and negative portions of the waveform. Figure ?? shows the output of the Class B Amplifier model on a 220 Hz sine wave, and Figure ?? shows the resulting harmonics generated.



(a) Class B output



(b) Class B harmonics

Figure 3: Class B Amplifier

2.10 Tube

The Tube is also based on research by Will Pirkle, specifically his analysis of real vacuum-tube behavior. In this case, the model represents a Class-A valve stage in which the tube conducts continuously throughout the waveform. This model also incorporates the same natural tube-style compression used in the Class-B amplifier model; however, it does not employ cascaded wave-shaping structures. This lightweight design makes the model ideal for stacking

multiple instances to build custom guitar preamp chains, if you are into that sort of thing. :)

Compared to the Class-B model, the tube model produces smoother, slightly darker, and more continuous saturation with fewer high-order harmonics, making it well suited for subtle drive and tone shaping.

Figure 4a shows the output of the Tube model on a 220 Hz sine wave, and Figure 4b shows the generated resulting harmonics.

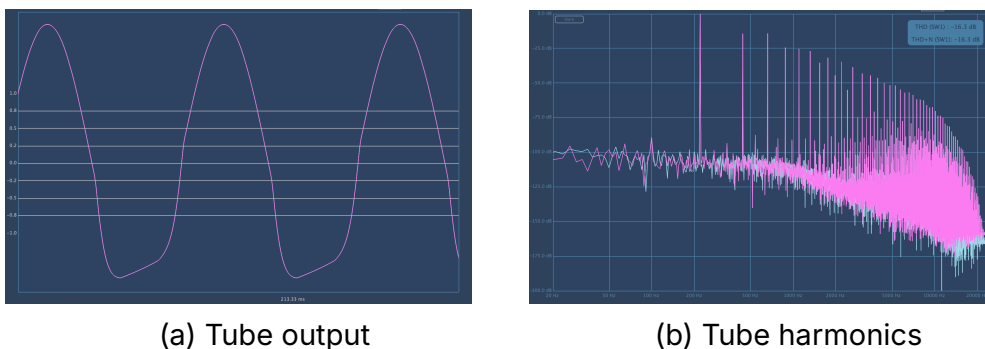


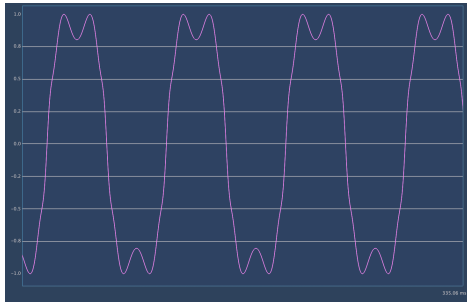
Figure 4: Tube

2.11 Circle Map

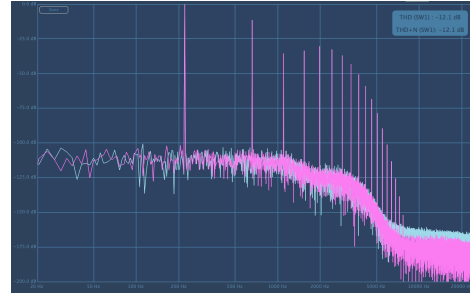
The Circle Map model is based on my own research on dynamical systems, which are mathematical systems defined by equations that evolve over time. The Circle Map is one such system and is particularly well suited for wave-shaping due to its one-dimensional structure, which maps an input value directly to an output while exhibiting rich nonlinear and chaotic behavior. [2].

The model also exhibits a natural compression effect, causing the transients to gently fold back into the signal as the drive increases. This compression helps maintain perceived loudness and smoothness while preventing harsh peaks, making the distortion feel controlled and musical rather than aggressive.

Figure 5a shows the output of the Circle Map model on a 220 Hz sine wave, and Figure 5b shows the generated resulting harmonics.



(a) Circle Map output



(b) Circle Map harmonics

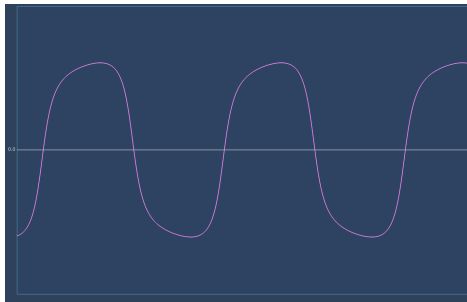
Figure 5: Circle Map

2.12 Tape

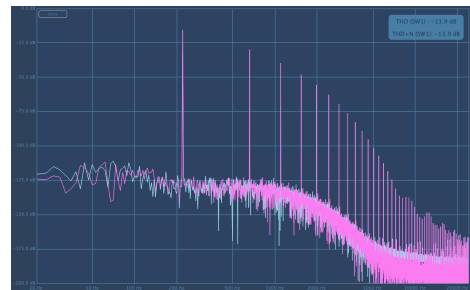
The Tape model emulates classic reel-to-reel behavior through smooth soft clipping, compression, and a characteristic low-shelf bump.

As drive increases, the waveform gradually rounds rather than clipping abruptly, causing peaks to compress smoothly while lower-level detail is brought forward. The low-shelf bump adds subtle low-frequency weight, helping the signal feel fuller and more cohesive without introducing harsh or brittle distortion.

Figure 6a shows the output of the Tape model on a 220 Hz sine wave, and Figure 6b shows the generated resulting harmonics.



(a) Tape output



(b) Tape harmonics

Figure 6: Tape

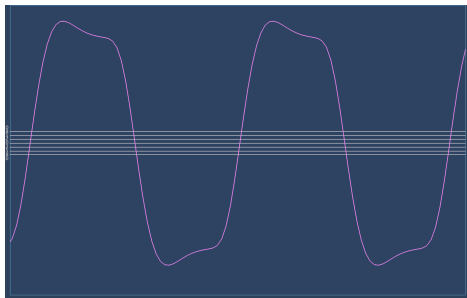
2.13 Overdrive

The Overdrive model loosely follows the behavior of classic mid-focused overdrive pedals, drawing inspiration from designs such as the Tube Screamer. Rather than producing heavy saturation, the distortion is intentionally soft and controlled, emphasizing the midrange while gently rolling off extreme low and high frequencies.

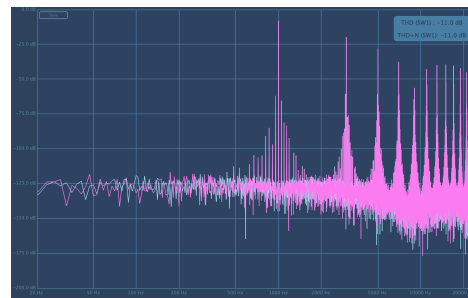
As drive increases, the signal is pushed into smooth, asymmetric clipping, resulting in a tight, focused overdrive that preserves note definition and pick attack. This makes the distortion feel responsive and articulate, particularly when used in front of an already-driven amplifier or amp model.

The Overdrive model is well suited for tightening low end, enhancing midrange presence, and pushing other distortion stages harder without overwhelming them. It works especially well on guitars for rhythm and lead tones, or as a tone-shaping stage placed before heavier distortion models.

Figure 7a shows the output of the Overdrive model on a 220 Hz sine wave, and Figure 7b shows the generated resulting harmonics.



(a) Overdrive output



(b) Overdrive harmonics

Figure 7: Overdrive

2.14 Bit Reducer

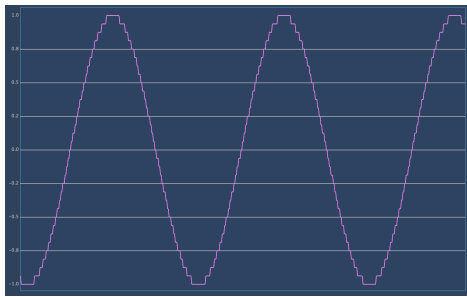
The Bit Reducer model reduces the effective bit depth of the incoming signal, introducing quantization distortion and stepped amplitude behavior. As resolution is lowered, the signal becomes increasingly

coarse and grainy, with added high-frequency artifacts characteristic of digital bit reduction.

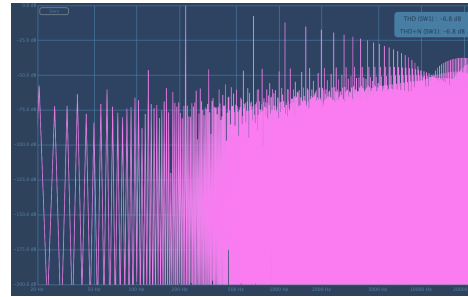
A hard clipper is applied at the output stage, and the Drive control pushes the bit-reduced signal into clipping as it is increased. This interaction allows the distortion to transition from subtle digital texture to aggressive, crushed saturation, depending on drive and depth settings.

Because the noise floor is inherently high with bit reduction, viewing the harmonics graph isn't that useful, but it's included anyway. :)

Figure 8a shows the output of the Bit Reducer model on a 220 Hz sine wave, and Figure 8b shows the generated resulting harmonics.



(a) Bit Reducer output



(b) Bit Reducer harmonics

Figure 8: Bit Reducer

3 Non-Controlled Features

To reduce static behavior and introduce subtle movement, Ripper incorporates several low-level dynamic processes. A small amount of pink noise is injected into the signal path at a fixed gain of approximately -70 dBFS, remaining well below the audible foreground while helping to de-correlate the processing over time.

In addition, the Tone control exhibits slow, continuous drift. Both the pivot frequency and the tilt gain gently move back and forth over time, causing gradual spectral variation rather than a fixed tonal balance. This keeps sustained material from sounding static or overly rigid.

The tilt gain drifts symmetrically in both directions, subtly shifting emphasis between low and high frequencies without changing the overall level or balance of the signal.

4 Header Controls

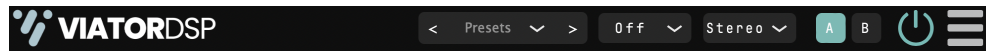


Figure 9: Header

The header area contains global controls for preset management, processing options, and plugin settings.

The Preset Manager allows presets to be saved, loaded, and selected directly from the interface. Users can also choose the preset storage location, making it easy to organize custom presets alongside factory content.

An Oversampling menu provides selectable processing rates of Off, 2×, 4×, 8×, and 16×. Higher oversampling values reduce aliasing and improve high-frequency clarity at the cost of increased CPU usage, allowing users to balance sound quality and performance as needed.

The Stereo Mode menu determines how stereo content is processed. Options include Stereo, Mid, and Sides. In Stereo mode, the full stereo signal is processed normally. In Mid or Sides mode, only the selected component is processed while the unprocessed counterpart is mixed back in—for example, selecting Mid processes the mid signal and then recombines it with the original sides, and vice versa.

An A-B Preset control allows quick comparison between two sets of parameter states, making it easy to audition changes without losing a previous setting. Every time a control is moved, all control's values are saved to the selected "A-B" state and toggling either state sets all controls to the saved values.

A Global Bypass control disables all processing while keeping the plugin loaded, enabling instant A/B comparison with the dry signal.

The Settings button opens a dedicated page containing additional resources and information, including links to Patreon, Gumroad, and

Discord, as well as a link to the user manual. This page also provides a shortcut for contacting support via email and includes the plugin's End User License Agreement (EULA).

5 Footer



Figure 10: Footer

The footer area contains global Input and Output controls, each paired with a small level meter for visual feedback.

The Input control adjusts the level feeding into the processing chain, allowing the signal to be driven more or less aggressively into the distortion stages. The accompanying input meter reflects the signal level entering the plugin.

The Output control sets the final level after all processing, making it easy to compensate for gain changes introduced by distortion or filtering. Its level meter displays the post-processing signal level, helping maintain consistent output and avoid unwanted clipping.

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References

- [1] W. Pirkle, "Addendum a19: Vacuum tube & distortion emulation, part 2," tech. rep., Music-DSP.org / Will Pirkle, 2019. Supplemental addendum to *Designing Audio Effect Plugins in C++*.
- [2] Wikipedia contributors, "Arnold tongue — standard circle map section." https://en.wikipedia.org/wiki/Arnold_tongue#Standard_circle_map, 2026. Accessed January 2026.